

```
In [1]: import numpy as np
```

```
In [2]: np.__version__
```

```
Out[2]: '2.1.3'
```

```
In [3]: import sys  
sys.version
```

```
Out[3]: '3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:37:03) [MSC v.192  
9 64 bit (AMD64)]'
```

Creating arrays

```
In [6]: my_list=[1,2,3,4,5]  
my_list,type(my_list)
```

```
Out[6]: ([1, 2, 3, 4, 5], list)
```

```
In [7]: arr=np.asarray(my_list)
```

```
In [8]: arr
```

```
Out[8]: array([1, 2, 3, 4, 5])
```

```
In [9]: type(arr)
```

```
Out[9]: numpy.ndarray
```

```
In [10]: np.arange(3.0),np.arange(10),np.arange(1,5)
```

```
Out[10]: (array([0., 1., 2.]),  
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9]),  
array([1, 2, 3, 4]))
```

```
In [12]: np.arange(10,10),np.arange(-10,4)
```

```
Out[12]: (array([], dtype=int64),  
array([-10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2,  
3]))
```

```
In [13]: np.arange(1,10,3)
```

```
Out[13]: array([1, 4, 7])
```

```
In [15]: np.zeros(3)
```

```
Out[15]: array([0., 0., 0.])
```

```
In [17]: np.zeros([2,2])
```

```
Out[17]: array([[0., 0.],  
[0., 0.]])
```

```
In [18]: np.zeros([3,3],dtype=int)
```

```
Out[18]: array([[0, 0, 0],
               [0, 0, 0],
               [0, 0, 0]])
```

```
In [19]: n = (6,7)
         n1 = (6,8)
         print(np.zeros(n1)),
         print(np.zeros(n)),

[[0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]]
[[0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0.]]
```

```
Out[19]: (None,)
```

```
In [20]: n,n1
```

```
Out[20]: ((6, 7), (6, 8))
```

```
In [25]: np.ones(3),np.ones([3,3])
```

```
Out[25]: (array([1., 1., 1.]),
         array([[1., 1., 1.],
               [1., 1., 1.],
               [1., 1., 1.]])
```

```
In [26]: np.random.rand(3)
```

```
Out[26]: array([0.21754029, 0.45853233, 0.35278106])
```

```
In [28]: np.random.rand(3,6)
```

```
Out[28]: array([[0.19304121, 0.61112371, 0.9615486 , 0.93739691, 0.81011366,
                0.36437398],
               [0.03645201, 0.65007723, 0.74153941, 0.60809133, 0.59943426,
                0.90480076],
               [0.78914171, 0.42519405, 0.07795575, 0.15109418, 0.38262857,
                0.61428263]])
```

```
In [29]: np.random.randint(3,6)
```

```
Out[29]: 4
```

```
In [32]: np.random.randint(3,10,2)
```

```
Out[32]: array([6, 8], dtype=int32)
```

```
In [33]: np.random.randint(3,10,(3,3))
```

```
Out[33]: array([[4, 7, 8],
               [4, 6, 8],
               [6, 5, 8]], dtype=int32)
```

```
In [34]: np.arange(1,13).reshape(3,4)
```

```
Out[34]: array([[ 1,  2,  3,  4],
               [ 5,  6,  7,  8],
               [ 9, 10, 11, 12]])
```

```
In [35]: np.arange(1,13).reshape(12, 1)
```

```
Out[35]: array([[ 1],
               [ 2],
               [ 3],
               [ 4],
               [ 5],
               [ 6],
               [ 7],
               [ 8],
               [ 9],
               [10],
               [11],
               [12]])
```

```
In [38]: b = np.random.randint(10,20,(5,4))
         b,type(b)
```

```
Out[38]: (array([[10, 19, 17, 11],
               [15, 19, 19, 17],
               [10, 17, 10, 18],
               [13, 15, 11, 14],
               [18, 14, 19, 17]], dtype=int32),
         numpy.ndarray)
```

```
In [39]: b[1:3]#slicing 1 to 2 row
```

```
Out[39]: array([[15, 19, 19, 17],
               [10, 17, 10, 18]], dtype=int32)
```

```
In [40]: b[1,2] #m,n element
```

```
Out[40]: np.int32(19)
```

```
In [41]: b[1,-1]
```

```
Out[41]: np.int32(17)
```

```
In [42]: b[0:-2] # : means matrix slicing
```

```
Out[42]: array([[10, 19, 17, 11],
               [15, 19, 19, 17],
               [10, 17, 10, 18]], dtype=int32)
```

operations

```
In [43]: a = np.random.randint(10,20,10)
         a
```

```
Out[43]: array([16, 11, 19, 16, 17, 14, 18, 10, 10, 16], dtype=int32)
```

```
In [44]: id(a)
```

```
Out[44]: 2107970264528
```

```
In [45]: arr
```

```
Out[45]: array([1, 2, 3, 4, 5])
```

```
In [46]: arr2 = np.random.randint(0,100,(10,10))  
arr2
```

```
Out[46]: array([[30, 39,  5, 11, 44, 65, 32, 25, 15, 54],  
                [ 6, 65, 63, 70, 14, 84,  9, 23, 10, 41],  
                [87, 84, 14, 23,  6, 54, 38, 17, 93, 93],  
                [98, 89, 45, 78, 11, 20, 27, 60, 55,  0],  
                [ 5, 50,  9, 11, 79, 49, 71, 21, 30, 22],  
                [56,  1, 73, 26, 98,  1, 78, 32, 69, 63],  
                [84, 22,  3, 24, 25, 13,  0, 39, 74, 27],  
                [76, 38, 94, 38, 15, 99, 21, 18, 97, 81],  
                [55, 76, 73,  1, 31, 55, 82, 73, 23, 40],  
                [40, 79, 84, 68, 52, 99, 11, 34, 40, 78]], dtype=int32)
```

```
In [47]: arr[:]
```

```
Out[47]: array([1, 2, 3, 4, 5])
```

```
In [48]: arr2[:]
```

```
Out[48]: array([[30, 39,  5, 11, 44, 65, 32, 25, 15, 54],  
                [ 6, 65, 63, 70, 14, 84,  9, 23, 10, 41],  
                [87, 84, 14, 23,  6, 54, 38, 17, 93, 93],  
                [98, 89, 45, 78, 11, 20, 27, 60, 55,  0],  
                [ 5, 50,  9, 11, 79, 49, 71, 21, 30, 22],  
                [56,  1, 73, 26, 98,  1, 78, 32, 69, 63],  
                [84, 22,  3, 24, 25, 13,  0, 39, 74, 27],  
                [76, 38, 94, 38, 15, 99, 21, 18, 97, 81],  
                [55, 76, 73,  1, 31, 55, 82, 73, 23, 40],  
                [40, 79, 84, 68, 52, 99, 11, 34, 40, 78]], dtype=int32)
```

```
In [49]: arr2[0:5]
```

```
Out[49]: array([[30, 39,  5, 11, 44, 65, 32, 25, 15, 54],  
                [ 6, 65, 63, 70, 14, 84,  9, 23, 10, 41],  
                [87, 84, 14, 23,  6, 54, 38, 17, 93, 93],  
                [98, 89, 45, 78, 11, 20, 27, 60, 55,  0],  
                [ 5, 50,  9, 11, 79, 49, 71, 21, 30, 22]], dtype=int32)
```

```
In [50]: arr2[1,4]
```

```
Out[50]: np.int32(14)
```

```
In [51]: arr2[-5,5]
```

```
Out[51]: np.int32(1)
```

```
In [52]: arr2
```

```
Out[52]: array([[30, 39, 5, 11, 44, 65, 32, 25, 15, 54],
               [ 6, 65, 63, 70, 14, 84, 9, 23, 10, 41],
               [87, 84, 14, 23, 6, 54, 38, 17, 93, 93],
               [98, 89, 45, 78, 11, 20, 27, 60, 55, 0],
               [ 5, 50, 9, 11, 79, 49, 71, 21, 30, 22],
               [56, 1, 73, 26, 98, 1, 78, 32, 69, 63],
               [84, 22, 3, 24, 25, 13, 0, 39, 74, 27],
               [76, 38, 94, 38, 15, 99, 21, 18, 97, 81],
               [55, 76, 73, 1, 31, 55, 82, 73, 23, 40],
               [40, 79, 84, 68, 52, 99, 11, 34, 40, 78]], dtype=int32)
```

```
In [53]: arr2[::-3] # from bottom gap -3
```

```
Out[53]: array([[40, 79, 84, 68, 52, 99, 11, 34, 40, 78],
               [84, 22, 3, 24, 25, 13, 0, 39, 74, 27],
               [98, 89, 45, 78, 11, 20, 27, 60, 55, 0],
               [30, 39, 5, 11, 44, 65, 32, 25, 15, 54]], dtype=int32)
```

```
In [54]: arr2[:-3]
```

```
Out[54]: array([[30, 39, 5, 11, 44, 65, 32, 25, 15, 54],
               [ 6, 65, 63, 70, 14, 84, 9, 23, 10, 41],
               [87, 84, 14, 23, 6, 54, 38, 17, 93, 93],
               [98, 89, 45, 78, 11, 20, 27, 60, 55, 0],
               [ 5, 50, 9, 11, 79, 49, 71, 21, 30, 22],
               [56, 1, 73, 26, 98, 1, 78, 32, 69, 63],
               [84, 22, 3, 24, 25, 13, 0, 39, 74, 27]], dtype=int32)
```

```
In [55]: arr2[-3:]
```

```
Out[55]: array([[76, 38, 94, 38, 15, 99, 21, 18, 97, 81],
               [55, 76, 73, 1, 31, 55, 82, 73, 23, 40],
               [40, 79, 84, 68, 52, 99, 11, 34, 40, 78]], dtype=int32)
```

```
In [56]: arr
```

```
Out[56]: array([1, 2, 3, 4, 5])
```

```
In [57]: arr.median()
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[57], line 1
----> 1 arr.median()

AttributeError: 'numpy.ndarray' object has no attribute 'median'
```

Without work on import* can you please find the median, mode)

```
In [61]: arr=np.arange(0,6)
arr
```

```
Out[61]: array([0, 1, 2, 3, 4, 5])
```

```
In [62]: arr.reshape(3,2)
```

```
Out[62]: array([[0, 1],
               [2, 3],
               [4, 5]])
```

```
In [64]: arr.reshape(2,3,order='C')
```

```
Out[64]: array([[0, 1, 2],
               [3, 4, 5]])
```

```
In [65]: arr.reshape(2,3,order='F') # print element with fortran
```

```
Out[65]: array([[0, 2, 4],
               [1, 3, 5]])
```

```
In [66]: arr.reshape(2,3,order='A') # A almost give you c type output
```

```
Out[66]: array([[0, 1, 2],
               [3, 4, 5]])
```

```
In [67]: mat = np.arange(0,100).reshape(10,10)
mat
```

```
Out[67]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
               [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
               [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
               [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [70]: row = 4
col = 5
mat[row,col]
```

```
Out[70]: np.int64(45)
```

```
In [71]: mat[4,5]
```

```
Out[71]: np.int64(45)
```

```
In [72]: mat[6] # befaulit it represent to rows
```

```
Out[72]: array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
In [73]: mat[:,col]
```

```
Out[73]: array([ 5, 15, 25, 35, 45, 55, 65, 75, 85, 95])
```

```
In [74]: mat[:,8]
```

```
Out[74]: array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [75]: mat[0:10:3]
```

```
Out[75]: array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [76]: mat[:, -3]
```

```
Out[76]: array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
               [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
               [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
               [ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9]])
```

```
In [77]: mat[2:6, 2:4] # 1:5 --> only row part /// 1:3 -- it indicates only column parts
```

```
Out[77]: array([[22, 23],
               [32, 33],
               [42, 43],
               [52, 53]])
```

```
In [78]: mat[1:2, 2:4]
```

```
Out[78]: array([[12, 13]])
```

masking

```
In [ ]:
```